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Using Contingency Contracting to Promote Social Interactions Among Students With ASD and Their Peers

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Abstract

The purpose of this study was to evaluate the effect of peer training implemented alone and the effect of combining contingency contracting with peer training on promoting social interactions among students with ASD and their peers. Three students with ASD and six typically developing peers enrolled in an inclusive elementary school participated in the study. Ten-minute observations were conducted during recess time to collect data on the participants' social interactions. The data obtained showed that peer training alone did not result in improvement in social interactions. However, upon the introduction of contingency contracting, which facilitated the use of prompting and reinforcement, the participants engaged in a significantly higher number of social interactions. The findings about the effect of peer training and contingency contracting were consistent across the participants. The study results suggest several implications for practice and directions for future research.

Keywords

autism, social interaction, social skills, peer-mediated interventions, contingency contracting

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Inclusive education has become a worldwide common educational practice in a movement to enhance socialization for students with disabilities (Camargo et al., 2014; Hullet, 2009; Rafferty et al., 2003; Wiener & Tardif, 2004). In spite of positive social outcomes for many students with disabilities that have been reported in the scientific literature on inclusive education, students with autism spectrum disorders (ASD) are far less likely to socially benefit from inclusive education than other students without receiving social interventions (Mesibov & Shea, 1996; Rogers, 2000).

One fact suggesting the need for systematic instruction to teach social skills to students with ASD relates to the profound deficits in social development and to the consequences of these deficits (Camargo et al., 2014). According to the *Diagnostic and Statistical Manual of Mental Disorders* (5th ed.; DSM-5; American Psychiatric Association [APA], 2013), individuals with ASD have persistent impairments in social communication and interaction. These impairments include difficulty in sharing social experiences with other people, establishing and maintaining relationships with peers, exhibiting social behaviors appropriate to different social contexts, and displaying verbal and nonverbal behaviors necessary to regulate social interactions (APA, 2013).

As defining characteristics of ASD, the deficits in social communication and social interaction are perhaps the most interfering characteristics that hamper development in different domains of functioning (APA, 2013). Indeed, if they remain untreated, these deficits contribute to poor academic achievement and limit the ability of many individuals with ASD to lead independent and fulfilling lives when they reach adulthood (Howlin et al., 2004; Welsh et al., 2001). These social impairments can also increase the likelihood for social withdrawal, anxiety, depression, obsessional disorders, tempers, suicidal thoughts, maladaptive behavior, and aggression (Bellini, 2006b; Charlop & Erickson, 2013; Green et al., 2000).

Although a vast number of interventions have been evaluated and discussed in the literature, a need still exists to design better interventions (Bellini, 2006a). A common trend found in the literature on teaching social skills to students with ASD is the recommendation to use interventions that incorporate elements and principles of applied behavior analysis (ABA), such as the use of prompting, modeling, and reinforcement (Reichow & Volkmar, 2010). This recommendation has been frequently mentioned in reviews of evidence-based practices for students with ASD such as those done by the National Autism Center (NAC) (2015) and the National Professional Development Center on Autism Spectrum Disorder (NPDC) (2014).

Incorporating ABA components, peer-mediated intervention (PMI) has become a paradigm of teaching social skills to students with ASD (Rogers,

2000). Supported by a tremendous amount of research on its effects, PMI has been one of the most recommended interventions for students with ASD (NAC, 2015; NPDC, 2014; Reichow & Volkmar, 2010). Reviews of the literature on PMI (e.g., Chan et al., 2009; Reichow & Volkmar, 2010) have suggested the effectiveness of different peer-mediated strategies in promoting a wide range of social behaviors and skills of students with ASD. Depending on the purpose and context of these interventions, PMI researchers incorporated different components into the interventions. However, the PMI strategies utilized in the studies all share similar features that include (a) recruiting peers who possess adequate social skills, (b) training the peers on their roles and responsibilities using different strategies such as modeling and role play, and (c) establishing opportunities to facilitate social interactions (Carter et al., 2012; Chan et al., 2009).

Although PMI has been shown to be effective in many studies, some literature on PMI has raised several concerns over the use of this type of instruction to teach social skills to students with ASD. A considerable body of research demonstrating the effects of PMI on improving social outcomes for children with ASD included other strategies that were incorporated into PMI. For example, in a recent study done by Hughes et al. (2013), the purpose was to evaluate the effect of PMI on promoting social interactions between three high school students with ASD and their peers. Hughes et al. (2013) conducted a series of training sessions that included teaching the peers to use appropriate ways to interact with the students with ASD. The training sessions also included teaching the peers to set social interaction goals and to self-monitor their progress toward achieving the goals. Although the intervention resulted in improvement in social interactions, it is impossible to determine the effects of PMI alone. In other words, the combination of PMI and self-management limited the conclusion that PMI in and of itself was effective. The same conclusion applies to other studies in which researchers combined other strategies such as adult prompting during peer interactions (Banda et al., 2010) and visual prompts and scripts (Harper et al., 2008; Owen-DeSchryver et al., 2008) with PMI.

Other research studies support the fact that PMI, when implemented alone, might have little to no effect on improving social skills for students with ASD. For example, Sainato et al., (1992) examined the effects of training peers to use social interaction strategies with three elementary school students with ASD. Sainato et al. (1992) found that peer training was not sufficient to enhance the use of the strategies. Therefore, the researchers trained the peers to use self-management techniques to evaluate their use of the social interaction strategies. The researchers found that the peers' use of the social interaction strategies improved only after the introduction of the

self-management intervention. Other studies support the same conclusion about the effectiveness of PMI when peer training was combined with visually based interventions.

In their research, Thiemann and Goldstein (2004) trained peers without disabilities to interact with students with ASD. According to the authors, the training slightly increased the frequency of some social skills for some of the students with a high level of prompting from adults. However, when the researchers trained the students with ASD to use written scripts that showed social phrases and statements, the interactions among the students improved further and the amount of adult prompting decreased. Similarly, Ganz et al. (2012) trained typically developing peers to interact with a student with ASD using visual scripts that included written social responses. Using a changing condition, multiple-baseline design, Ganz et al. (2012) stated that the treatment package (i.e., peer training and visual script) was more effective than the peer training alone in promoting social interactions between the students.

In addition to these findings, there are other practical issues concerning PMI. First, many students with ASD have difficulty appreciating social reinforcers (e.g., a smile or compliment) from peers, even if the peers receive training (Taylor, 2013). On the other hand, peers, regardless of the amount of training they get, may not find interacting with students with ASD a pleasant experience due to the lack or inappropriateness of social responses from students with ASD (Carter et al., 2012; DiSalvo & Oswald, 2002). Second, although PMI has been used to reduce the amount of instruction and guidance from adults, interventions involving peers still require adult guidance to prompt students to interact and to monitor their progress (Carter et al., 2012). Adult guidance and feedback may be considered atypical in everyday interaction among children (Crick & Dodge, 1994).

To address these concerns, some literature suggests the use of tangible reinforcement and pairing the reinforcement with social consequences in order to make social interactions more motivating and enjoyable for both students with ASD and their peers (Taylor, 2013). Some literature also recommends combining PMI with strategies that allow students with ASD and their peers to prompt themselves to interact with one another and to evaluate their social performance (Ladd & Mize, 1983; Weiss & Harris, 2001). One intervention that has the potential for addressing these concerns and for improving social interactions between students with ASD and their peers is contingency contracting, also referred to as behavioral contracting, (Coope et al., 2007).

According to Cooper et al. (2007), a contingency contract is a written document that involves at least two parties (e.g., the teacher and the student)

and that shows the skill or behavior of interest, the conditions under which the skill or behavior should be performed, and the type and amount of reinforcement the client will receive. Combining contingency contracting with PMI may improve the effectiveness of the intervention. Every person involved in the contract verbalizes and writes rules in which a specified behavior (i.e., a task) leads to a specified consequence (i.e., a reward; Cooper et al., 2007). Because the delivery of rewards indicated in the contract is usually delayed, a person for whom the contract is developed self-manages a targeted behavior in order for that person to get a reward (Cooper et al., 2007). In addition, because a contingency contract is a permanent product that is reviewed on a regular basis, the contract serves as a prompt for the individual to keep engaging in targeted behaviors (Cooper et al., 2007). Moreover, according to Maag (2004), because students involved in a contract have the opportunity to negotiate and agree on the components of the contract, the students are likely to have a sense of ownership of their behaviors. That sense of ownership ensures active student participation and more adherence to rules specified in the contract (Maag, 2004).

In spite of the features and possible effects of contingency contracting on social outcomes for student with ASD and their peers, the amount of research on the use of contingency contracting with students with ASD is relatively small (Bowman-Perrott et al., 2015). Moreover, no research to date has examined the effectiveness of combining peer training with contingency contracting in improving social interactions. Furthermore, although social development of children with ASD is the most critical determinant of future adjustment and prognosis for these children (as cited in Charlop & Erickson, 2013) and therefore should be the main focus of their education, little, if any, research on social skills interventions for students on the autism spectrum has been conducted in Saudi Arabia. The Saudi Ministry of Education has placed emphasis on inclusive education by encouraging schools to provide learning environments where students with disabilities can socially interact with their general education peers (Ministry of Education, 2001). Nevertheless, in practice inclusive schools in Saudi Arabia do not show strict adherence to the special education legislation with regard to placing students with disabilities in supportive learning environments (Alquraini, 2011). This lack of adherence may raise a concern over teachers' awareness of the importance of improving social skills in students with ASD and over their knowledge about implementing effective social skills interventions (Alquraini, 2011).

The current study is the first intervention study that focuses on social interactions among students with ASD and their typically developing peers in Saudi Arabia. The primary purpose of this study was to expand the research base on PMI and on contingency contracting in two ways. First, the study was

conducted to evaluate the effect of training peers on improving social interactions among the peers and students with ASD. Second, the study examined the effect of combining peer training with contingency contracting on the students' social interactions.

Method

Participants and Setting

Participants. Three boys diagnosed with ASD participated in the study and received the interventions implemented in the study. Fahad was 9 years old and was diagnosed with ASD based on the Gilliam Autism Rating scale ([GARS], Gilliam, 1995). GARS showed that the severity of Fahad's ASD was within the high level. The school records indicated that Fahad had an IQ of 59 based on the Stanford-Binet Intelligence Scale (5th edition) and that he had language disorders and hyperactivity. In terms of his social skills, Fahad avoided interacting with peers during his non-instructional time; instead, he engaged in solitary play. As for his academic skills, Fahad was unable to read and write and exhibited difficulty in learning elementary mathematics. Saud was 11 years old and was diagnosed with ASD based on GARS. The severity level of the diagnosis was moderate. Saud had an IQ of 78 based on the Stanford-Binet Intelligence Scale (5th edition), and he demonstrated adequate literacy and language skills. Although Saud was able to speak, he rarely interacted with other students and lacked motivation to communicate with his peers. Sattam was 8 years old and had moderate ASD based on GARS. The school psychologist indicated that he was unable to administer an intelligence test for Sattam because of Sattam's high level of repetitive behaviors. Sattam had language disorders and limited academic and social skills. The students with ASD engaged in a social program where they had opportunities to interact with trained peers. However, the school principle stated that the program was of little effect.

In addition to the students with ASD, six boys without disabilities participated in the study. The age of the nondisabled peers in the study ranged from 11 to 12 years. The students were sixth graders, had been nominated by the school as an example of what other students should be, and were above average in terms of their academic skills. Some of the peers had previously participated in social skills programs where they were prompted by teachers to interact with students with ASD. The authors obtained written consent from the students' parents and oral assent from the students before starting the study.

Setting. The study took a place in an inclusive elementary school located in the west side of Riyadh, Saudi Arabia. The school had a population of about

600 students and 45 teachers, ten of whom were special education teachers. The school had two special education programs designed to educate students with ASD and students with ADHD. About 20 students with ASD attended the school and received special education in self-contained classrooms. The self-contained classrooms met the criteria set by the Ministry of Education in Saudi Arabia. They included a maximum of five students and were equipped with materials necessary to provide effective individualized and small-group instruction such as small round activity tables. The students with ASD had an opportunity to spend recess time with their general education peers. However, due to the large number of students attending the school, the students with ASD were placed in a special recess area. This area included two large playgrounds; each was about 32 feet long and 20 feet wide. Each playground had a variety of play materials and equipment such as swing sets and soccer balls and goals. Every school day, a small number of general education students, about 20 students, were allowed to enter the playground and spend time with the students with ASD.

Experimental Design

The design of the study was an ABC multiple-baseline design across three participants (Alberto & Troutman, 2013; Baer et al., 1968), in which A, B, and C designate the baseline, peer training alone, and peer training with contingency contracting, respectively. Because there were three tiers and three different start points for each intervention condition, the design would demonstrate the effects of peer training implemented alone and the effects of peer training combined with contingency contracting at three different points in time (Cooper et al., 2007; Horner et al., 2005). The number of demonstrations would be sufficient to establish a functional relation between each intervention and the social behaviors of the participants (Kazdin, 2011). Peer training was implemented for the first participant when baseline data collected during 4–5 sessions revealed a stable or descending trend in the target skills. Prior to implementing peer training for the second participant, the authors waited until a clear trend in the skills of the first participant was evident in at least 4–5 sessions. The researchers repeated the same procedure with the third participant. The introduction of contingency contracting with peer training followed the same procedures.

Dependent Measures

The researchers conducted 10-min observations to collect data on the target behaviors. The number of observations in each phase was at least five. They

used event recording to collect data on the peers' social initiations and on the social responses from the students with ASD. Event recording is recommended to collect information about behaviors that have a clear beginning and ending point and that occur at a steady rate such as most social behaviors (Cooper et al., 2007; Kazdin, 2011; Maag, 2004).

Definition of peer initiations. The dependent variables in the study were social initiations by the peers and social responses to social initiations by the students with ASD. A social initiation is operationally defined as a communicative behavior that meets the following criteria. First, the behavior takes any form of complete verbalization, vocalization, or gesture. If two or more communicative behaviors occur simultaneously (e.g., saying "hi" while waving hands), the behaviors are counted as one response. Second, the behavior triggers a response from a listener with ASD. That is, the behavior should be exhibited in a way that elicits a response from the student with ASD. Examples are asking an initial question, asking a question that pertains to a response from a student with ASD, or a greeting. Third, the behavior is clearly directed toward a listener with ASD. A social initiation is considered clearly directed to a listener with ASD if the speaker initiates while looking at, turning his head toward, orienting his body toward, or standing or sitting within three feet of the student with ASD. If one of the school's faculty or staff encourages or prompts the peer to interact with the student with ASD, the behavior will not be considered an initiation.

Definition of appropriate social responses. A social response to a peer's social initiation is operationally defined as a communicative behavior that meets the following criteria. First, the behavior takes any form of complete verbalization, vocalization, or gesture. If two or more communicative behaviors occur simultaneously (e.g., saying "wow" while raising eyebrows), the behaviors are counted as one response. Second, the behavior is a contingent response. That is, the behavior should be exhibited within 8 s of the peer's initiation toward the student with ASD. Examples of a contingent response are answering a question or responding to a request. Third, the behavior is clearly directed toward a peer who initiates a social interaction. A response is considered clearly directed to a peer if the student with ASD gives the response while looking at, turning his head toward, orienting his body toward, or standing or sitting within three feet of the peer giving the initiation. A response that is directed to a peer other than the one initiating is not counted as a social response. If the student with ASD responds to one of the school's faculty or staff, the behavior will not be considered a social response. In addition, if one of the school's faculty or staff encourages or

prompts the student with ASD to interact with peers, the behavior will not be considered a social response.

Reliability

To evaluate the reliability of data, two observers independently and simultaneously recorded the data in each condition. Special education teachers who had expertise in data collection served as the reliability coder, and the reliability coders and the first author sat within 8–10 feet of the participants when observing them. The inter-observer agreement was calculated using the total agreement method by dividing the smaller frequency by the larger frequency and multiplying by 100. The reliability levels for each phase are mentioned below. The researchers also plotted the data recorded by the secondary observer in Figures 1 and 2 because of the large variability of the reliability data.

The total inter-observer agreement regarding the baseline phase for Fahad and his peers was evaluated during 40% of the sessions. For Saud and his peers, the agreement was measured during 33% of the sessions and during 30% of the sessions for Sattam and his peers. In the peer training condition, the reliability for Fahad and his peers was calculated during 60% of the sessions, in 40% of the sessions for Saud and his peers, and in 33% of the sessions for Sattam and his peers. In the contingency contracting phase, the agreement for Fahad and his peers was evaluated during 41% of the sessions. Regarding Saud, Sattam, and their peers the agreement was collected during 40% of the sessions. Table 1 indicates the range and mean of the total inter-observer agreement for each participant and in each condition.

Experimental Conditions

Baseline. Prior to the implementation of peer training and contingency contracting, the observers collected baseline data following the procedures described under the Experimental Design section of this study. The collection of baseline data took place during 10-min probe periods of recess time, which lasted for about 30 min. During the baseline phase, the participants did what they typically do during recess time. There were opportunities for the participants to interact with one another, and there were some prompts directed to the participants by teachers to encourage the students to interact. As stated previously, the observers sat or stood within 8–10 feet of the participants. When asked by the students about their presence at the recess area, the observers indicated that they were observing how students play during recess.

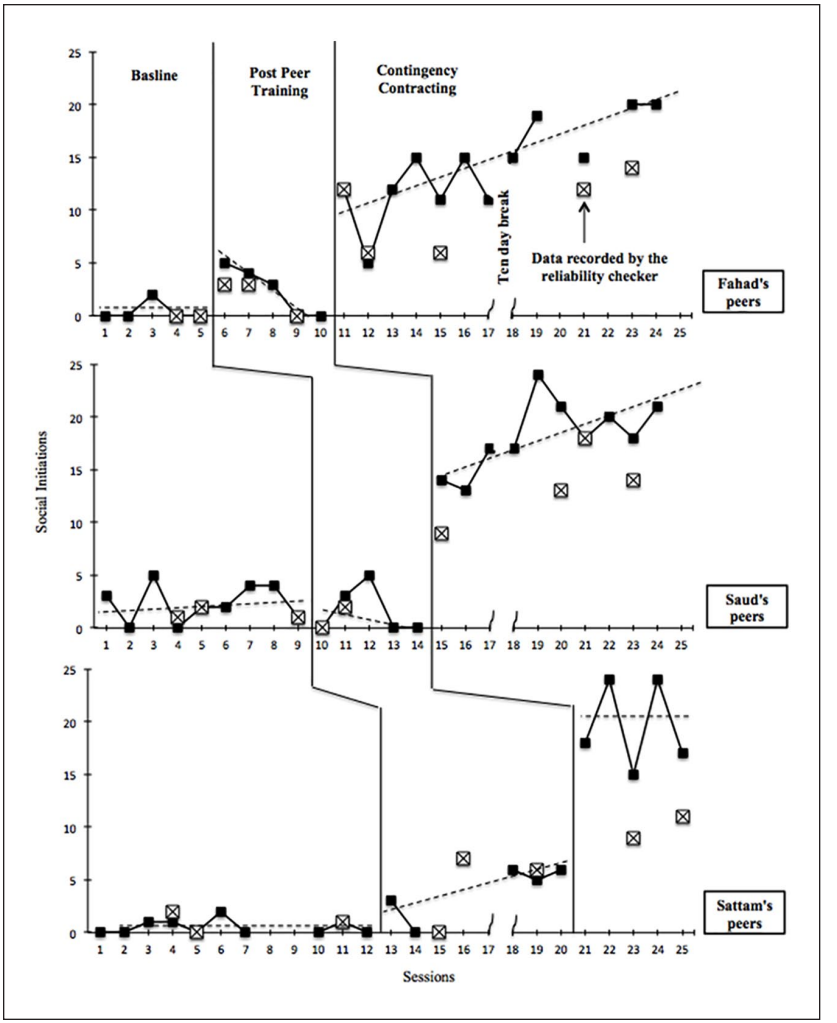


Figure 1. Frequency of social behaviors initiated by the peers across the three phases.

Peer training alone. The researchers conducted 3–4 sessions of peer training immediately after the last session of the baseline; each session lasted for approximately 30 min. There was no data collection during the peer training. Peer training involved teaching the typically developing peers to work with the students with ASD. In the first session, the first author explained and

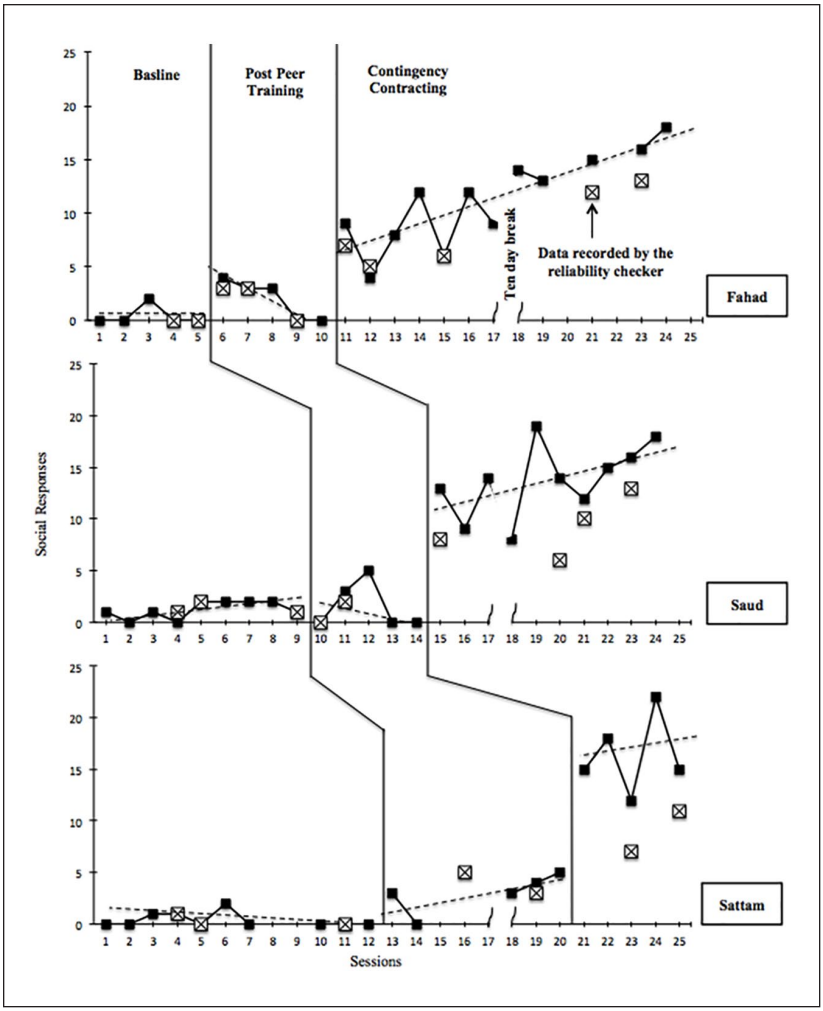


Figure 2. Frequency of social responses to the peers' initiations across the three phases.

discussed the characteristics of students with ASD with their peers. The discussion included topics such as impairments and deficits that students with ASD have, the needs of students' with ASD, and how one can help them meet these needs. The discussion ended with answering questions the students had. In the following sessions, the first author modeled how and when the students

Table 1. The Ranges and Means of the Total Inter-observer Agreement.

	Baseline		Peer training		Contingency contracting	
Student	Range (%)	Mean (%)	Range (%)	Mean (%)	Range (%)	Mean (%)
Fahad	100	100	60–100	85	55–100	80
Saud	0–100	66	67–100	83	43–100	71
Sattam	50–100	91	75–100	89	58–73	64

should prompt, model, and provide corrective feedback to or praise the students with ASD. Finally, the students and the first author acted out how to interact with the students with ASD, and the first author pretended to be the student with ASD (Chan et al., 2009). The first author continued conducting the training sessions until the peers were able to implement the teaching strategies (i.e., prompting, modeling, and providing feedback) appropriately. After the last session of peer training, the researchers started data collection similar to the baseline condition.

Peer training and contingency contracting. During this phase, the authors implemented the contingency contracting along with peer training. The researchers created a contingency contract for each student with ASD and for each peer. They followed the steps for creating a behavior contract as recommended by Walker and Shea (1984) with each student. In a small classroom, the first author sat with the student around a small table and followed the step shown in Table 2. The task assigned to the students with ASD was to interact with peers, and the task assigned to their peers was to help the student with ASD interact by prompting, modeling, and providing feedback about social behaviors. The researcher also told every student that they would meet every week to renegotiate the contract and to provide the reward. He also told the student that they would meet every day to review the contract and the task record included in the contract. The observers collected data on the dependent variables after writing the contract and did not interact with the students unless a student displayed self-injurious behavior or aggressive behavior directed toward other students.

Intervention Fidelity

The researchers collected data on intervention fidelity using three checklists. The first checklist included important steps for selecting and training peers to work with students with ASD suggested by Neitzel (2008). The researchers

Table 2. Steps for Creating a Contingency Contract.

Step	Procedure
1	The researcher establishes and maintains rapport with the student.
2	The researcher explains the purpose of the meeting.
3	The researcher explains the meaning of a contract.
4	The researcher gives examples of a contract.
5	The researcher discusses the task assigned to the student.
6	The researcher and student agree on tasks.
7	The researcher discusses possible reinforcers with student.
8	The researcher negotiates the ratio of the task to the reinforcer with the student.
9	The researcher identifies the time to get the reinforcer with the student.
10	The researcher sets the date for renegotiation.
11	The researcher prints a copy for the student.
12	The researcher reads the contract with the student.
13	The researcher gives verbal affirmation and elicits affirmation from student.
14	The researcher signs the contract and has the student sign it.
15	The researcher congratulates the student.

used the checklist in every training session. The second checklist included the steps for creating contingency contracts as suggested by Walker and Shea (1984) and was used during the creation of contingency contracts. The last checklist was used to collect information on the fidelity for 20% of the weekly renegotiation sessions. The intervention fidelity for peer training was collected during all training sessions and was 100% for all sessions. The fidelity for the creation of the contracts was measured for every participant except for Fahad's peers because the researchers were unable to find a teacher to observe the session and collect the needed information. The intervention fidelity regarding the development of the contract was 100% for Saud, Sattam, and their peers. The fidelity regarding the weekly renegotiation sessions was evaluated during at least 20% of the sessions, and the integrity for renegotiating the contract was 100% for every student and in each session.

Results

The researchers analyzed the data using single case research design visual analysis. Visually analyzing data in single case research has an advantage over the use of statistical analyses in helping researchers identify interventions that only result in profound effects, a highly desirable kind of effects in applied research and settings (Baer, 1977; Kazdin, 2011).

Table 3. Numbers of the Peers' Social Initiations.

	Baseline		Peer training		Contingency contracting	
Student	Range	Mean	Range	Mean	Range	Mean
Fahad	0–2	0.4	0–4	2.4	5–20	14
Saud	0–5	2.33	0–5	1.6	13–24	18.3
Sattam	0–2	0.5	0–7	4.5	15–24	19.6

Peers' Social Initiations

Figure 1 presents the frequency of social behaviors initiated by the peers across the three phases, and Table 3 shows the range and mean of the peers' social initiation throughout the study. There were two peers assigned to work with every student with ASD. The following sections provide a description of the results about the performance of Fahad's, Saud's, and Sattam's peers.

Fahad's peers. During the baseline phase, the data pattern of social initiations from Fahad's peers was stable without a trend. The peers exhibited no social initiations in the first and last two sessions of the phase. After the implementation of peer training, the frequency of social initiations increased. However, because the peers did not display any social initiation in the last two sessions of the peer training phase, the data pattern in this phase showed a descending trend. Upon the introduction of contingency contracting, Fahad's peers immediately increased the social initiations, in comparison to the peer training condition, and the data pattern in this phase showed an ascending trend.

Saud's peers. The baseline frequency of social initiations by Saud's peers indicates a slightly increasing trend. The frequency ranged from 0 to 5 with a mean of 2.33, which reveals little variability in the data. After receiving training, the peers initiated social behaviors without noticeable improvement in the frequency of the initiations, compared to the baseline. There was overlap between the baseline and peer training data, and the trend of the social behaviors was decreasing. When the contingency contracting was conducted, the peers showed improvement in the frequency of social initiations with no overlap with the baseline and peer training data. The immediacy of change in the behaviors was evident, given the large difference in the frequency of social initiations between the three phases. The trend of the behaviors during the contingency contracting phase was significantly ascending.

Sattam's peers. The data path was flat with no trend and with little variability during the baseline. After receiving training, the peers did not immediately increase the number of social initiations, with little overlap in the baseline and peer training data. The data pattern in this phase was slightly divergent, and the trend of the data was increasing. Although the trend of the peer training data was increasing, the start of the contingency contracting led to a rapid change in the level of social initiations. There was a flat line with no trend during the contingency contracting phase.

Students' with ASD Social Responses

Figure 2 displays the data on the frequency of social responses from the students with ASD across the baseline, peer training, and contingency contracting phases. Table 4 shows the range and mean of the students' social responses throughout the study. The following sections provide a description of the results about the performance of the students with ASD.

Fahad. Fahad had few opportunities to respond to social initiations during the baseline phase. The baseline data for Fahad were stable. When his peers received training, the level of social responses for Fahad increased during the first 3 sessions of the peer training phase. However, the peers stopped initiating in the last 2 sessions. As a result, the trend of Fahad's social behaviors was descending. When contingency contracting was implemented, Fahad and his peers engaged in more social interactions. Thus, Fahad had a higher number of social responses compared to the numbers in the baseline and peer training phases. The change of Fahad's social responses was immediate as he exhibited a total number of 21 responses at the beginning of the contingency contracting phase, compared to two social responses in the last three sessions of the peer training phase. The contingency contracting data showed an ascending trend and did not overlap with data obtained in previous phases.

Saud. There were a small number of opportunities for Saud to respond during the baseline phase. The range and mean of the responses suggest a relatively stable data pattern with no variability. When his peers received training, Saud started to respond more frequently as his peers initiated more social behaviors. However, his responses decreased at the end of the phase, making the data overlap with the previous phase. The range of the responses in this phase was 0–5 with a mean of 1.6, indicating little variability in the data. There was a descending trend for Saud's social responses in this phase. Upon the introduction of contingency contracting, Saud started to

Table 4. Numbers of the Students' with ASD Correct Responses.

	Baseline		Peer training		Contingency contracting	
Student	Range	Mean	Range	Mean	Range	Mean
Fahad	0–2	0.4	0–4	2	4–18	11.33
Saud	0–2	1.2	0–5	1.6	8–19	14.6
Sattam	0–2	0.4	0–5	3.3	12–22	16.4

have higher social initiations, leading him to respond more frequently. In fact, the improvement of Saud's social behaviors was noticeable as he exhibited 36 social responses in the first three sessions of the contingency contracting phase. The contingency contracting data did not overlap with data collected in the other phases. The data in this phase showed an ascending trend and had a moderate level of variability.

Sattam. Sattam's rate of social responses was small during the baseline, and the baseline data showed a slightly decreasing trend with no variability. During the peer training condition, he had more opportunities to have social interactions with peers. Therefore, his social behaviors slightly improved. The trend of the peer training data was ascending, and the data had little overlap with the baseline data. The changes in the social responses for Sattam were noticeable when contingency contracting was implemented. His responses during the contingency contracting condition significantly improved. The data had a moderate level of variability and did not overlap with data obtained in the baseline and peer training phases. The data pattern showed an ascending trend during this phase.

Discussion

Research Focus, Purpose, and Results

The current study extended the research on PMI by demonstrating the effect of peer training that was implemented with and without contingency contracting on promoting social interactions between students with ASD and their peers. The results showed that peer training without adult prompting and reinforcement did not result in noticeable improvement in social interactions among the participants. However, when the participants signed their contracts, which helped to provide them with prompts and reinforcement, they engaged in a significantly higher number of interactions. The increase in the frequency of the peers' social initiations improved the number of social

responses from the students with ASD. That is, the students with ASD were able to respond to more frequent social initiations.

The study findings are consistent with results obtained in other research that show PMI in and of itself may be essential but insufficient to improve social interactions between students with ASD and their peers (Ganz et al., 2012; McConnell, 2002; Sainato et al., 1992; Thiemann & Goldstein, 2004). In this research, including the current study, peer training did not significantly improve social initiations from the typically developing peers, and in turn, it did not promote social interactions with the students with ASD. There are several reasons that may be responsible for the limited effects when peer training was implemented alone. First, according to Crick and Dodge (1994), social interaction is a circulating process that starts with encoding and interpreting social cues and ends with enacting a social behavior. Crick and Dodge (1994) stated that children evaluate the outcomes of their social behaviors in a given context, and based on the outcomes, they decide whether to perform the same behaviors in similar social contexts. The lack of improvement in the peers' social performance during the peer training phase could be explained by the lack of social cues (e.g., responses from the students with ASD or adult prompting) or by undesired social outcomes of the peers' social initiations.

Second, literature on ASD shows that many children with ASD do not appreciate social reinforcers (e.g., a smile or compliment) from peers, even if the peers have training (Taylor, 2013). Moreover, the literature reveals that peers, regardless of the amount of training they get, may not find interacting with students with ASD a pleasant and motivating experience due to the lack or inappropriateness of social responses from students with ASD (Carter et al., 2012; DiSalvo & Oswald, 2002). Therefore, the literature suggests that the use of tangible rewards during social training and instruction is of great benefit as it can help to address the lack of social motivation for both students with ASD and their peers (Taylor, 2013).

In order to capitalize on the benefits of PMI, the researchers implemented contingency contracting in the study because it serves as a visual prompt and facilitates the use of reinforcement (Cooper et al., 2007). The peer initiations improved when the PMI was combined with reinforcement and prompting provided via contingency contracting. Ultimately, the students with ASD benefited from the increased initiations because they had more opportunities to engage with the peers as demonstrated in the study. These findings are consistent with other research suggesting the importance of using prompts and reinforcement (e.g., Harper et al., 2008; Owen-DeSchryver et al., 2008) in maximizing the effect of interventions designed to enhance social interaction between students with ASD and their peers. The results are also consistent with research conclusions that PMI must

include additional components in order to be effective (e.g., Sainato et al., 1992; Thiemann & Goldstein, 2004).

Limitations and Directions for Future Research

The current study has several limitations. The first limitation relates to the reliability of the study data. Low levels of reliability were evident in multiple sessions in the baseline because of the extremely low frequency of target behaviors (Thorndike & Thorndike-Crist, 2010). Low reliability was also noticed in some sessions of the contingency contracting phase possibly due to the distractions (e.g., noise) associated with the high levels of interaction, which could make observation more difficult (Repp et al., 1988). As shown in Figures 1 and 2, there was some variability between the study data and the data collected by the reliability checker in some sessions. In spite of the variability, the overall data pattern suggested a functional relation between contingency contracting and the dependent measures. Although low reliability averages are not unusual in research focusing on social skills (e.g., Banda et al., 2010; Kohler et al., 1995), the reliability averages were lower than the minimum acceptable value of reliability averages (i.e., 0.80; Kratochwill et al., 2010).

The second limitation relates to treatment fidelity and to social validity. There was no information on treatment fidelity for Fahad's peers because the authors could not find another person to observe the first author while he was signing the contract with the peers. So there was no way to determine the accuracy and conformity of implementing the contracts (Smith et al., 2007). As a result, it is difficult to determine the accuracy of delivering the intervention. A related limitation is that the authors did not collect social validity data because they had to finish the study earlier than it was planned; the Ministry of Education asked school districts to start final examinations earlier than it was officially determined. Without such information, it is difficult to determine the social validity of the intervention with regard to the goals, effects, and acceptability of the intervention procedures (Gresham & Lopez, 1996). Additional research is needed to evaluate the social importance of the intervention effects and the acceptability of the intervention.

The third limitation pertains to the generalizability of the study findings. In comparison to typical students with ASD, the three students who participated in the study had higher social skills. Therefore, the study results may apply only to students with ASD who have already acquired social skills but who need to be provided with opportunities to perform the acquired skills. In addition to these limitations, the data collection for some of the participants

was not continuous due to the students' repeated absences from school and to a ten-day school vacation. Collecting continuous data is a key feature of the research design and is essential for reaching valid conclusions about intervention effects (Kazdin, 2011). The patterns of the participants' social interactions might have been different if the researchers had collected the data continuously (Kazdin, 2011).

Furthermore, because of time constraints, the researchers were unable to evaluate the effect of the study interventions on generalization and maintenance of the social behaviors. Therefore, it is difficult to determine the effect of the interventions on the participants' abilities to maintain and generalize social interactions in new and different contexts. Because an ultimate goal of applied research is to evaluate interventions that result in enduring improvement without having subjects continue receiving the interventions (Baer et al., 1968), the lack of information about generalization and maintenance may limit the conclusion about the intervention effects. Therefore, future research is required to determine the effect of peer training and contingency contracting on maintaining and generalizing social interactions. Finally, because the study did not focus on the qualitative aspects of social interaction, future research is needed to examine the effects of PMI on the overall quality of relationship such as the ability to maintain the relationship between students with ASD and their peers.

Practical Implications

The results of the study suggest that educators should be aware that placing students with ASD in inclusive settings does not necessarily promote social integration (Rogers, 2000). In fact, students with ASD need specially designed interventions in order to socially benefit from inclusive education (Mesibov & Shea, 1996). Crucial to planning effective social interventions is the consideration of modifying the social environment by facilitating opportunities where students with ASD can successfully interact with their general education peers. Creating a supportive social environment requires providing special instruction to both students with ASD and their peers.

Because students with ASD often exhibit aberrant behaviors, their peers may have negative attitudes toward them, thus avoiding interacting with them (DiSalvo & Oswald, 2002). Therefore, it is important to educate general education students about the characteristics of ASD and to teach them how to appropriately interact with students with ASD. It is also important to remember that in order to maximize the effectiveness of peer training, teachers should use other interventions alongside peer training to help general education peers maintain interaction with students with ASD.

In addition to the importance of training and prompting peers, teachers should be aware that students with ASD and their peers might lose interest in maintaining social interactions. Students with ASD usually show no motivation to interact with peers, and as a result, the peers may not find the interactions enjoyable, thus avoiding further interactions (Taylor, 2013). Students with ASD may also be less interested in social interaction due to the continuous interaction failure and to the difficulty to understand others' communicative behaviors. Therefore, to enhance the effectiveness of a social skills intervention, students with ASD and their peers should be given the opportunity to experience successful and enjoyable social interactions (Taylor, 2013). Teachers can provide tangible rewards at initial phases of the intervention to students with ASD and their peers to make the interaction enjoyable, thus increasing the level of the students' social motivation (Taylor, 2013).

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References

Alberto, P. A., & Troutman, A. C. (2013). *Applied behavior analysis for teachers* (9th ed.). Person Education, Inc.

Alquraini, T. (2011). Special education in Saudi Arabia: Challenges, perspectives, future possibilities. *International Journal of Special Education*, *26*(2), 149–159.

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders*. American Psychiatric Publishing. http://doi.org/10.1176/appi.books.9780890425596

Baer, D. M. (1977). "Perhaps it would be better not to know everything". *Journal of Applied Behavior Analysis*, *10*(1), 167–172. http://dx.doi.org/10.1901/jaba.1977.10-167

Baer, D. M., Wolf, M. M., & Risley, T. R. (1968). Some current dimensions of applied behavior analysis. *Journal of Applied Behavior Analysis*, *1*(1), 91–97. https://doi.org/10.1901/jaba.1968.1-91

Banda, D. R., Hart, S. L., & Liu-Gitz, L. (2010). Impact of training peers and children with autism on social skills during center time activities in inclusive

classrooms. *Research in Autism Spectrum Disorders*, *4*(4), 619–625. http://dx.doi.org/10.1016/j.rasd.2009.12.005

Bellini, S. (2006a). *Building social relationships: A systematic approach to teaching social interaction skills to children with autism spectrum disorders and other social difficulties*. Autism Asperger Publishing Co.

Bellini, S. (2006b). The development of social anxiety in adolescents with autism spectrum disorders. *Focus on Autism and Other Developmental Disabilities*, *21*(3), 138–145. http://dx.doi.org/10.1177/10883576060210030201

Bowman-Perrott, L., Burke, M. D., de Marin, S., Zhang, N., & Davis, H. (2015). A meta-analysis of single-case research on behavior contracts: Effects on behavioral and academic outcomes among children and youth. *Behavior Modification*, *39*(2), 247–269. https://doi.org/10.1177/0145445514551383

Camargo, S. P. H., Rispoli, M., Ganz, J., Hong, E. R., Davis, H., & Mason, R. (2014). A review of the quality of behaviorally-based intervention research to improve social interaction skills of children with ASD in inclusive settings. *Journal of Autism and Developmental Disorders*, *44*, 2096–2116. http://dx.doi.org/10.1007/s10803-014-2060-7

Carter, E. W., Sisco, L. G., & Chung, Y. (2012). Peer-mediated support interventions. In R. McCauley & P. A. Prelock (Eds.), *Treatment of autism spectrum disorders: Evidence-based intervention strategies for communication and social interactions* (pp. 221–254). Paul H. Brookes Publishing Co.

Chan, J. M., Lang., R., Rispoli, M., O'Reilly, M., Sigafoos, J., & Cole, H. (2009). Use of peer-mediated interventions in the treatment of autism spectrum disorders: A systematic review. *Research in Autism Spectrum Disorders*, *3*(4), 876–889. https://doi.org/10.1016/j.rasd.2009.04.003

Charlop, M., & Erickson, M. (2013). An evidence-based social skills group for children with autism. In A. Bondy & M. Weiss (Eds.), *Teaching social skills to people with autism: Best practices in individualized interventions* (pp. 109–124). Woodbine House, Inc.

Cooper, J. O., Heron, T. E., & Heward, W. L. (2007). *Applied behavior analysis* (2nd ed.). Person Education, Inc.

Crick, N. R., & Dodge, K. A. (1994). A review and reformulation of social information-processing mechanisms in children's social adjustment. *Psychological Bulletin*, *115*(1), 74–101. http://doi.org/10.1037/0033-2909.115.1.74

DiSalvo, C. A., & Oswald, D. P. (2002). Peer-mediated interventions to increase the social interaction of children with autism consideration of peer expectancies. *Focus on Autism and Other Developmental Disabilities*, *17*(4), 198–207. https://doi.org/10.1177/10883576020170040201

Ganz, J. B., Heath, A. K., Lund, E. M., Camargo, S. P. H., Rispoli, M. J., Boles, M., & Plaisance, L. (2012). Effects of peer-mediated implementation of visual scripts in middle school. *Behavior Modification*, *36*(3), 378–398. http://dx.doi.org/10.1177/0145445512442214

Gilliam, J. (1995). *Gilliam Autism Rating Scale*. Pro-Ed.

Green, J., Gilchrist, A., Burton, D., & Cox, A. (2000). Social and psychiatric functioning in adolescents with Asperger syndrome compared with conduct disorder. *Journal of Autism and Developmental Disorders*, *30*(4), 279–293. https://doi.org/10.1023/A:1005523232106

Gresham, F. M., & Lopez, M. F. (1996). Social validation: A unifying concept for school-based consultation research and practice. *School Psychology Quarterly*, *11*(3), 204–227. http://doi.org/10.1037/h0088930

Harper, C. B., Symon, J. B. G., & Frea, W. D. (2008). Recess is time-in: Using peers to improve social skills of children with autism. *Journal of Autism and Developmental Disorders*, *38*(5), 815–826. http://dx.doi.org/10.1007/s10803-007-0449-2

Horner, R. H., Carr, E. G., Halle, J., Mcgee, G., Odom, S., & Wolery, M. (2005). The use of single-subject research to identify evidence-based practice in special education. *Exceptional Children*, *71*(2), 165–197. http://dx.doi.org/10.1177/001440290507100203

Howlin, P., Goode, S., Hutton, J., & Rutter, M. (2004). Adult outcome for children with autism. *Journal of Child Psychology and Psychiatry*, *45*(2), 212–229. https://doi.org/10.1111/j.1469-7610.2004.00215.x

Hughes, C., Harvey, M., Cosgriff, J., Reilly, C., Heilingoetter, J., Brigham, N., ... & Bernstein, R. (2013). A peer-delivered social interaction intervention for high school students with autism. *Research and Practice for Persons with Severe Disabilities*, *38*(1), 1–16. http://dx.doi.org/10.2511/027494813807046999

Hullet, K. E. (2009). *Legal aspects of special education*. Pearson Education, Inc.

Kazdin, A. E. (2011). *Single-case research designs: Methods for clinical and applied settings* (2nd ed.). Oxford University Press, Inc.

Kohler, F. W., Strain, P. S., Hoyson, M., Davis, L., Donina, W. M., & Rapp, N. (1995). Using a group-oriented contingency to increase social interactions between children with autism and their peers: A preliminary analysis of corollary supportive behaviors. *Behavior Modification*, *19*(1), 10–32. http://dx.doi.org/10.1177/01454455950191002

Kratochwill, T., Hitchcock, J. H., Horner, R. H., Levin, J. R., Odom, S. L., Rindskopf, D. M., & Shadish, W. R. (2010). *Single-case designs technical documentation*. What Works Clearinghouse. https://ies.ed.gov/ncee/wwc/Document/229

Krebs, M. L., Mcdaniel, D. M., & Neeley, R. A. (2010). The effects of peer training on the social interactions of children with autism spectrum disorders. *Education*, *131*(2), 393–403.

Ladd, G. W., & Mize, J. (1983). A cognitive-social learning model of social-skill training. *Psychological Review*, *90*(2), 127–157. http://dx.doi.org/10.1037/0033-295X.90.2.127

Maag, J. W. (2004). *Behavior management: From theoretical implications to practical applications* (2nd ed.). Wadsworth/Thomson Learning.

McConnell, S. R. (2002). Interventions to facilitate social interaction for young children with autism: Review of available research and recommendations

for educational intervention and future research. *Journal of Autism and Developmental Disorders*, *32*(5), 351–372. https://link.springer.com/article/10.1023/A:1020537805154

Mesibov, G. B., & Shea, V. (1996). Full inclusion and students with autism. *Journal of Autism and Developmental Disorders*, *26*(3), 337–346. https://doi.org/10.1007/BF02172478

Ministry of Education of Saudi Arabia. (2001). *Regulations of special education programs and Institutes of Saudi Arabia*. https://departments.moe.gov.sa/SPED/Documents/serules.pdf

National Autism Center. (2015). *Findings and conclusions: National standards project, phase 2: Addressing the need for evidence-based practice guideline for autism spectrum disorder*. National Autism Center.

National Professional Development Center on Autism Spectrum Disorder. (2014). *Evidence-based practices*. http://autismpdc.fpg.unc.edu/evidence-based-practices

Neitzel, J. (2008). *Steps for implementation: PMII for elementary, middle, and high school*. The National Professional Center on Autism Spectrum Disorders, Frank Porter Graham Child Development Institute, The University of North Carolina.

Owen-DeSchryver, J. S., Carr, E. G., Cale, S. I., & Blakeley-Smith. (2008). Promoting social interactions between students with autism spectrum disorders and their peers in inclusive school settings. *Focus on Autism and Other Developmental Disorders*, *23*(1), 15–28. http://dx.doi.org/10.1177/1088357608314370

Rafferty, Y., Piscitelli, V., & Boettcher, C. (2003). The impact of inclusion on language development and social competence among preschoolers with disabilities. *Exceptional Children*, *69*(4), 467–479. https://doi.org/10.1177/001440290306900405

Reichow, B., & Volkmar, F. R. (2010). Social skills interventions for individuals with autism: Evaluation for evidence-based practices within a best evidence synthesis framework. *Journal of Autism and Developmental Disorders*, *40*(2), 149–166. https://doi.org/10.1007/s10803-009-0842-0

Repp, A. C., Nieminen, G. S., Olinger, E., & Brusca, R. (1988). Direct observation: Factors affecting the accuracy of observers. *Exceptional Children*, *55*(1), 29–36.

Rogers, S. J. (2000). Interventions that facilitate socialization in children with Autism. *Journal of Autism and Developmental Disorders*, *30*(5), 399–409. https://doi.org/10.1023/A:1005543321840

Sainato, D. M., Goldstein, H., & Strain, P. S. (1992). Effects of self-evaluation on preschool children's use of social interaction strategies with their classmates with autism. *Journal of Applied Behavior Analysis*, *25*(1), 127–141. http://dx.doi.org/10.1901/jaba.1992.25-127

Smith, S. W., Daunic, A. P., & Taylor, G. G. (2007). Treatment fidelity in applied educational research: Expanding the adoption and application of measures to ensure evidence-based practice. *Education and Treatment of Children*, *30*(4), 121–134. http://dx.doi.org/10.1353/etc.2007.0033

Taylor, B. (2013). Improving the social behavior of children with autism. In A. Bondy & M. Weiss (Eds.), *Teaching social skills to people with autism: Best practices in individualized interventions* (pp. 1–24). Woodbine House, Inc.

Thiemann, K. S., & Goldstein, H. (2004). Effects of peer training and written text cueing on social communication of school-age children with pervasive developmental disorder. *Journal of Speech, Language, and Hearing Research*, *47*(1), 126–144. http://dx.doi.org/10.1044/1092-4388(2004/012)

Thorndike, R., & Thorndike-Christ, T. (2010). *Measurement and evaluation in psychology and education* (8th ed.). Pearson Education, Inc.

Walker, J., & Shea, T. (1984). *Behavior management: A practical approach for educators* (3rd ed.). Times Mirror/Mosby College Publishing.

Weiss, M. J., & Harris, S. L. (2001). Teaching social skills to people with autism. *Behavior Modification*, *25*(5), 758–802. https://doi.org/10.1177/0145445501255007

Welsh, M., Parke, R. D., Widaman, K., & O'Neil, R. (2001). Linkages between children's social and academic competence: A longitudinal analysis. *Journal of School Psychology*, *39*(6), 463–481. https://doi.org/10.1016/S0022-4405(01)00084-X

Wiener, J., & Tardif-Williams, C. Y. (2004). Social and emotional functioning of children with learning disabilities: Does special education placement make a difference? *Learning Disabilities Research and Practice*, *19*(1), 20–32. http://dx.doi.org/10.1111/j.1540-5826.2004.00086.x

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